



Computational Analysis of mRNA Stability and Modifications: Melting Simulations and Free Energy Perturbation Studies



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ABSTRACT

- This study performed **molecular dynamics simulations** of the **A-riboswitch-adenine complex** (PDB ID: **1y26**), contains a metabolite-sensing mRNA sequence, from **300–400K** to examine **thermal denaturation**. Analyses of **Root-Mean-Square Deviation**, **Radius of gyration**, **hydrogen bonding**, and **Watson–Crick base pairing** revealed progressive destabilization and partial melting at high temperatures.
- This study provides atomistic insight into RNA melting under stress and outlines a computational framework that can be extended to **RNA modifications** and **free energy perturbation (FEP)** in future work.

RESULTS

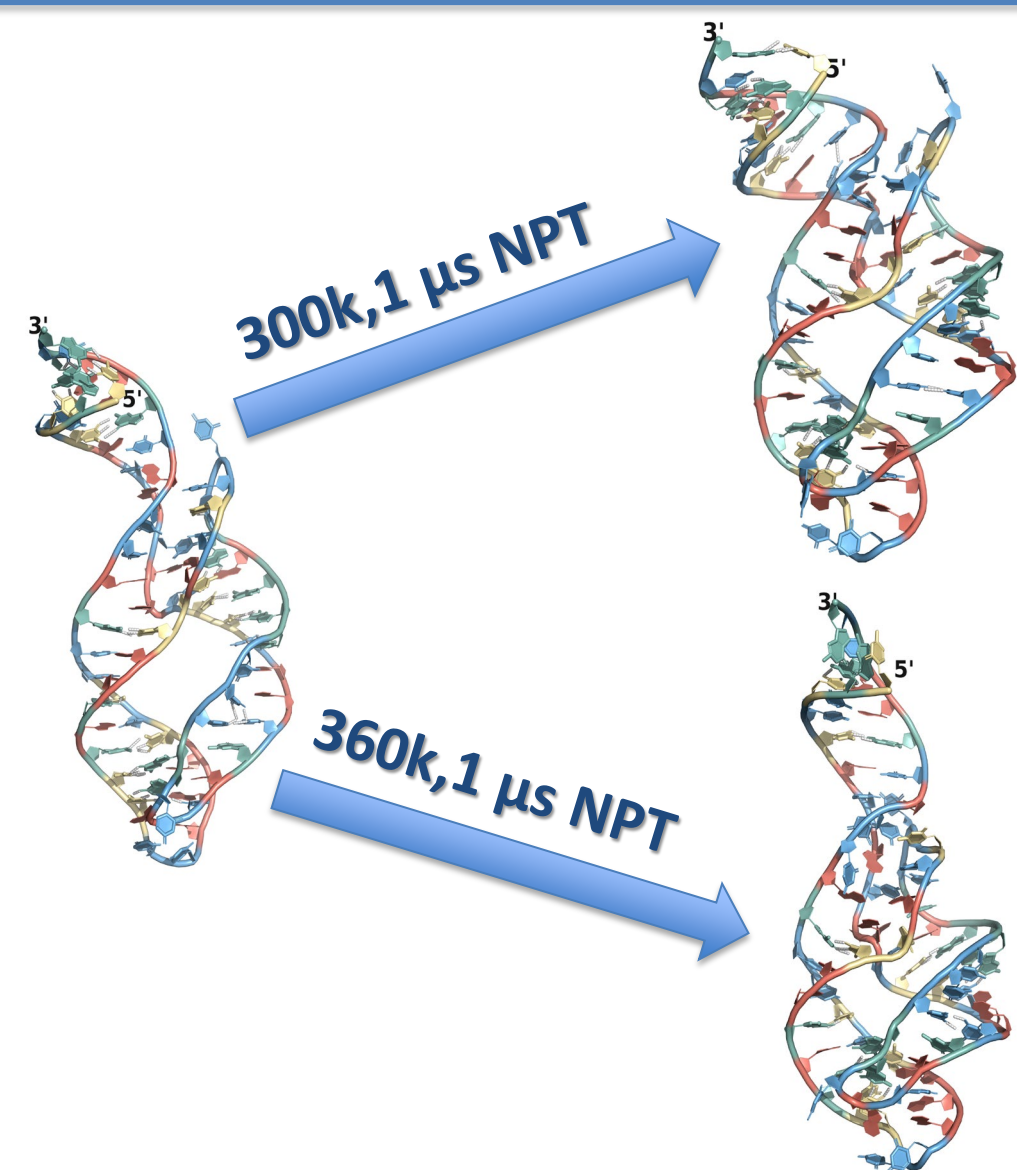


Fig 1. Structure snapshots of the native 1Y26 backbone and after 1 μ s NPT MD simulation at 300 K and 360 K.

- 1y26 maintains a **relatively stable** tertiary structure at **300K** after simulation, whereas the simulation at **360K** shows a significant structural deviations with **partial melting** of the tertiary structure.

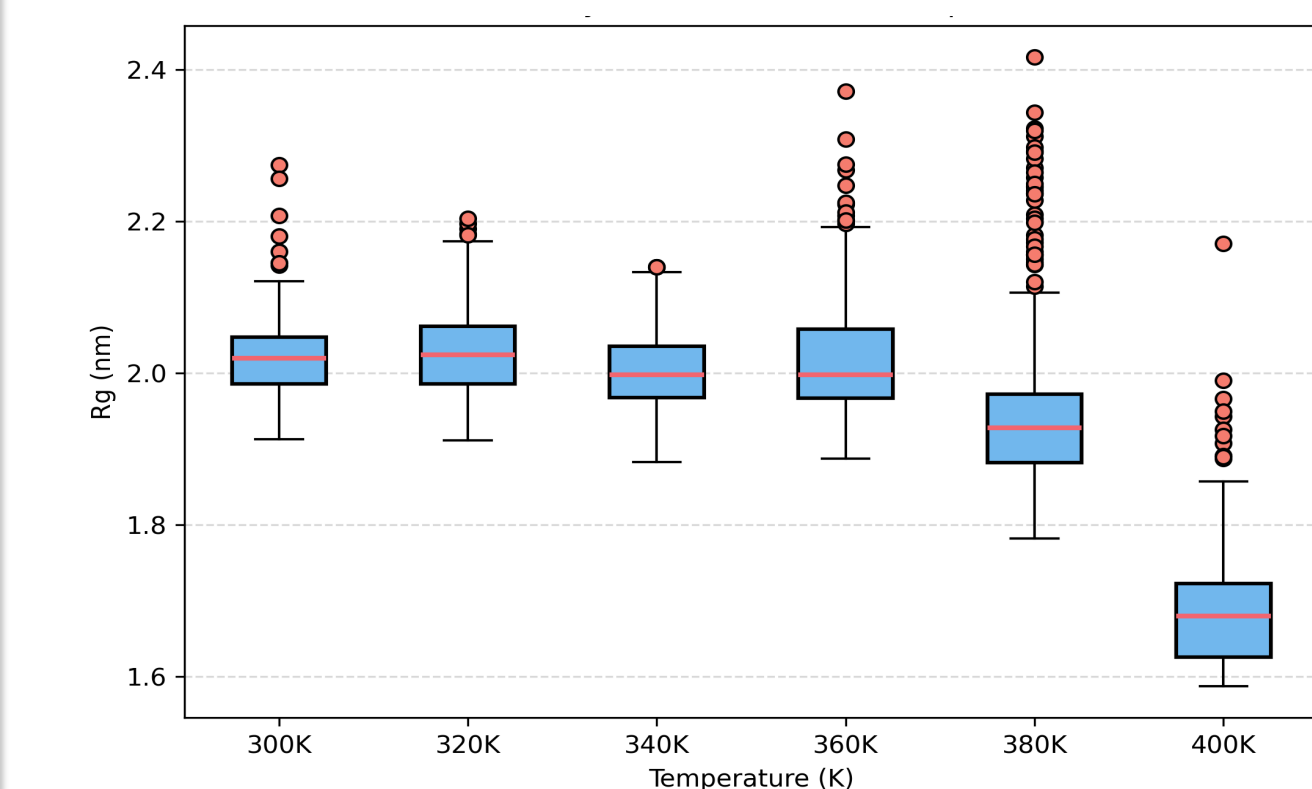
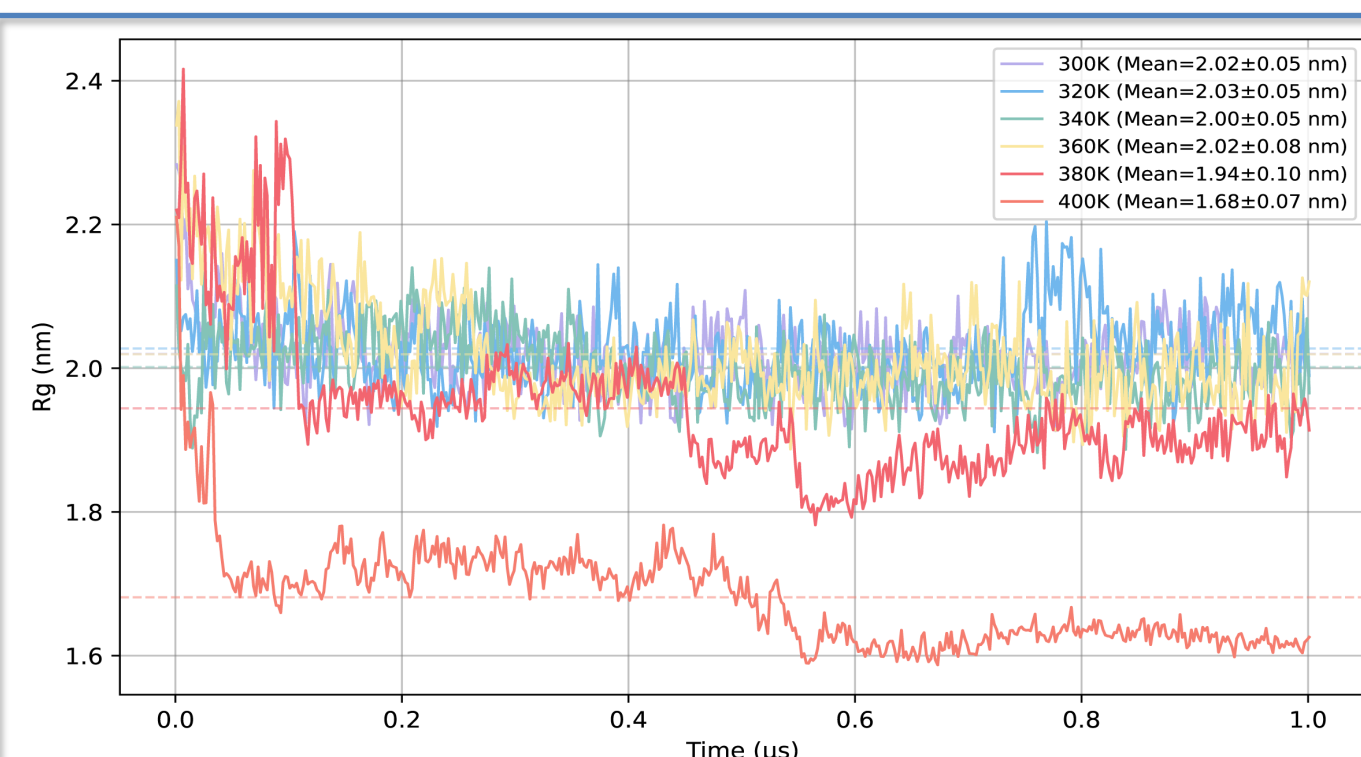


Fig 3.1. Temporal evolution of Rg for 1Y26 at 300–400 K over 1 μ s.

Fig 3.2. Box-and-whisker distributions of Rg for 1Y26 at 300–400 K over 1 μ s.

- The **Rg** shows only **minor fluctuations** at **lower temperatures**, while a **significant decrease** is observed at **elevated temperatures**, particularly at 400K (**exceed the boiling point of water**). This indicates a more **compact RNA conformation** rather than unfolding, which may suggest **temperature-induced structural collapse**.

CONCLUSION

- As expected, RNA stability is **temperature-dependent**, with **elevated temperatures** are more **disruptive** to the secondary and tertiary structure, as reflected by **increases in RMSD** and **decreases in the number of WcC pairs**. At very **high temperatures**, a **reduction in the Rg** is observed, which may represent a **simulation artifact** since such conditions exceed the boiling point of water.
- Future investigations should therefore focus on the **300–360 K** range to enable more meaningful comparison with experimental data.

INTRODUCTION

- RNA stability** governs folding, interactions, and biological function, yet its atomistic response to heat remains difficult to capture experimentally. **Molecular dynamics (MD)** simulations provide a powerful approach to probe thermal effects on RNA.
- This study investigated the 1y26 RNA across 300–400K, combining **NVT** and **NPT** phases to assess temperature-driven conformational changes. Analyses of Root-Mean-Square Deviation, Radius of gyration, hydrogen bonding, and Watson–Crick base pairs show **progressive destabilization** and **partial melting** at elevated temperatures.
- These results highlight melting pathways and establish a computational framework for predicting RNA stability under stress, while also pointing toward future extensions involving **RNA modifications** and **free energy perturbation (FEP)**.

METHODOLOGY

- Initial structure:** 1y26
- Processing**
Equilibration(NVT) + Production(NPT)
duration: 1000 ps + 1 μ s
temperature: 300K- 400K; pressure: 1 bar(NPT)
- Quantitative Metrics**
Structural drift: Root-Mean-Square Deviation
Global stability: Radius of gyration
Local dynamics: Base-pair occupancy
- Computational Details**
Software: GROMACS 2023.2-gcc-9.5.0-jzxesel

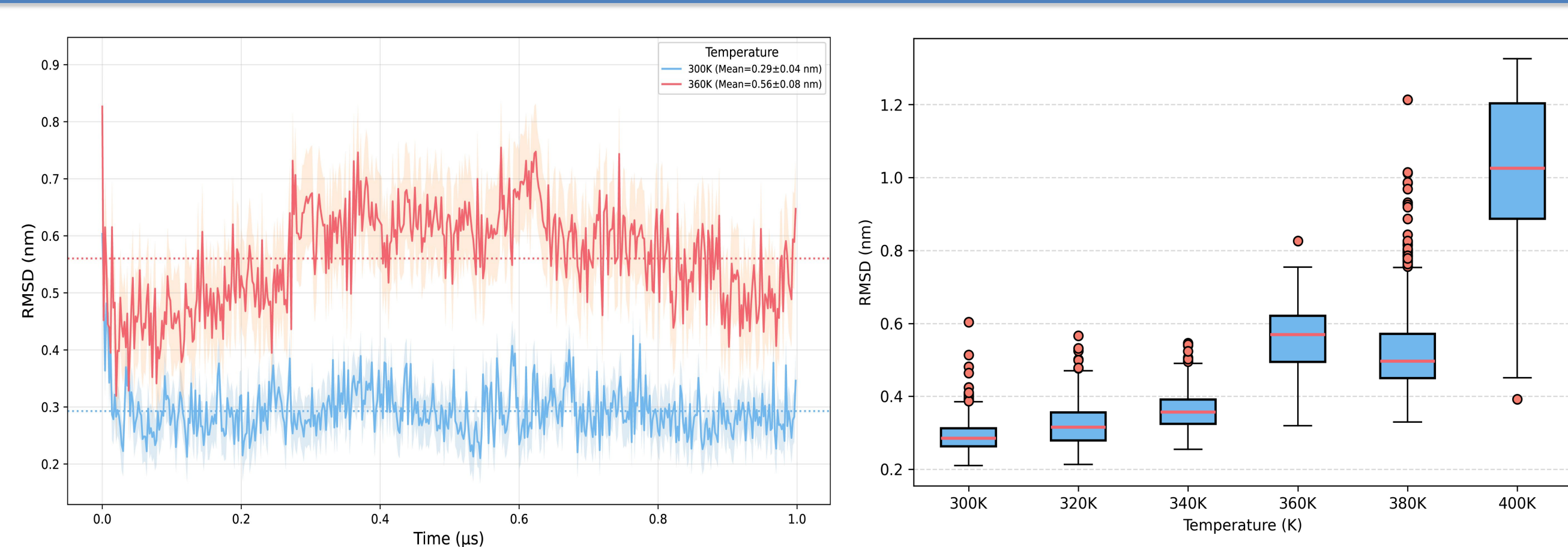


Fig 2.1. Fig Temporal evolution of RMSD for 1Y26 at 300 K and 360 K over 1 μ s. RMSD is calculated from the last frame of NVT.

Fig 2.2. Box-and-whisker distributions of RMSD for 1Y26 at 300–400 K over 1 μ s.

- The **RMSD values increase with temperature**, representing a trend of **higher structural deviations** at elevated temperatures. **Significant differences** are observed between low (300K) and high (360–400K) temperatures, indicating **reduced structural stability** under thermal stress.

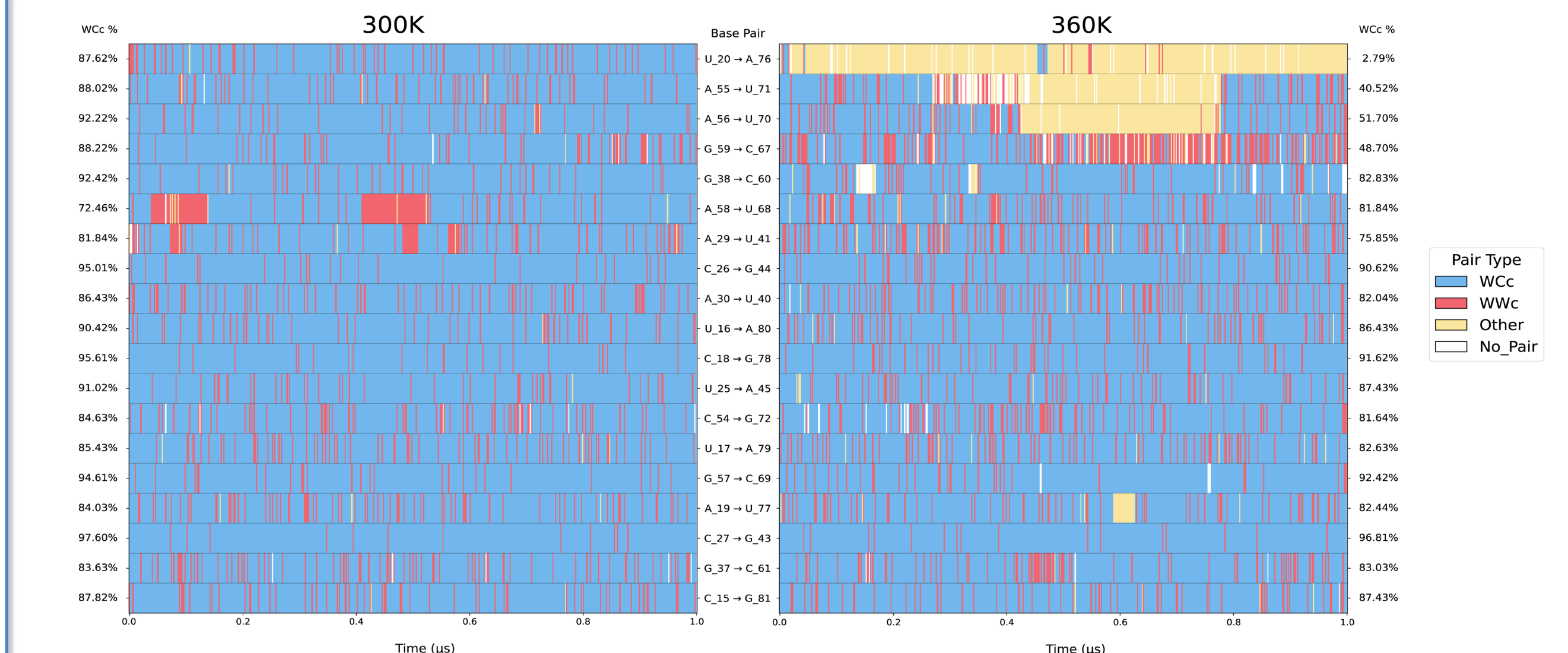


Fig 4.1. State maps for the 15 base pairs with the largest WcC frequency discrepancy for 1Y26 at 300 K and 360 K over 1 μ s. Base pairs are classified as WcC, WWc, Other, or No_Pair.

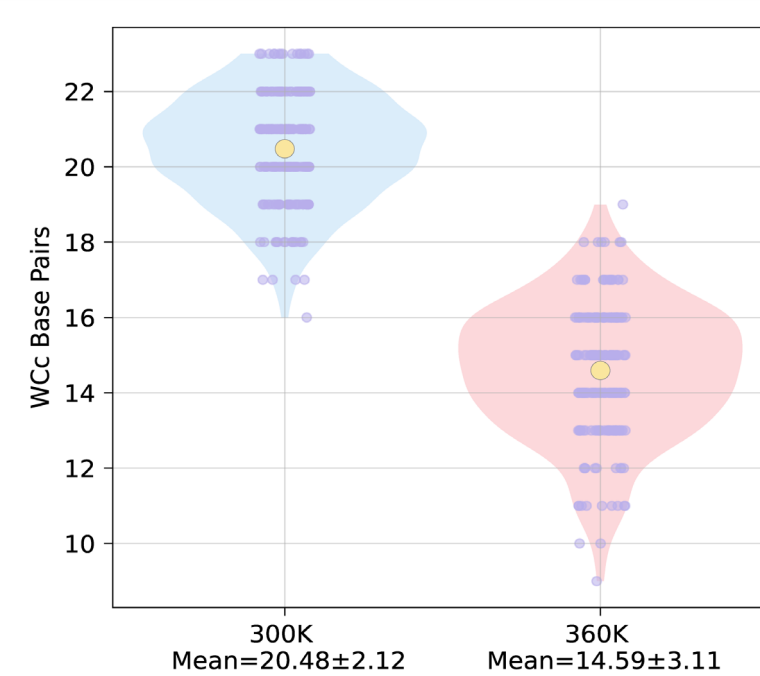


Fig 4.2. Distribution pattern of WcC base-pair counts for 1Y26 at 300 K and 360 K over 0.6–1 μ s.

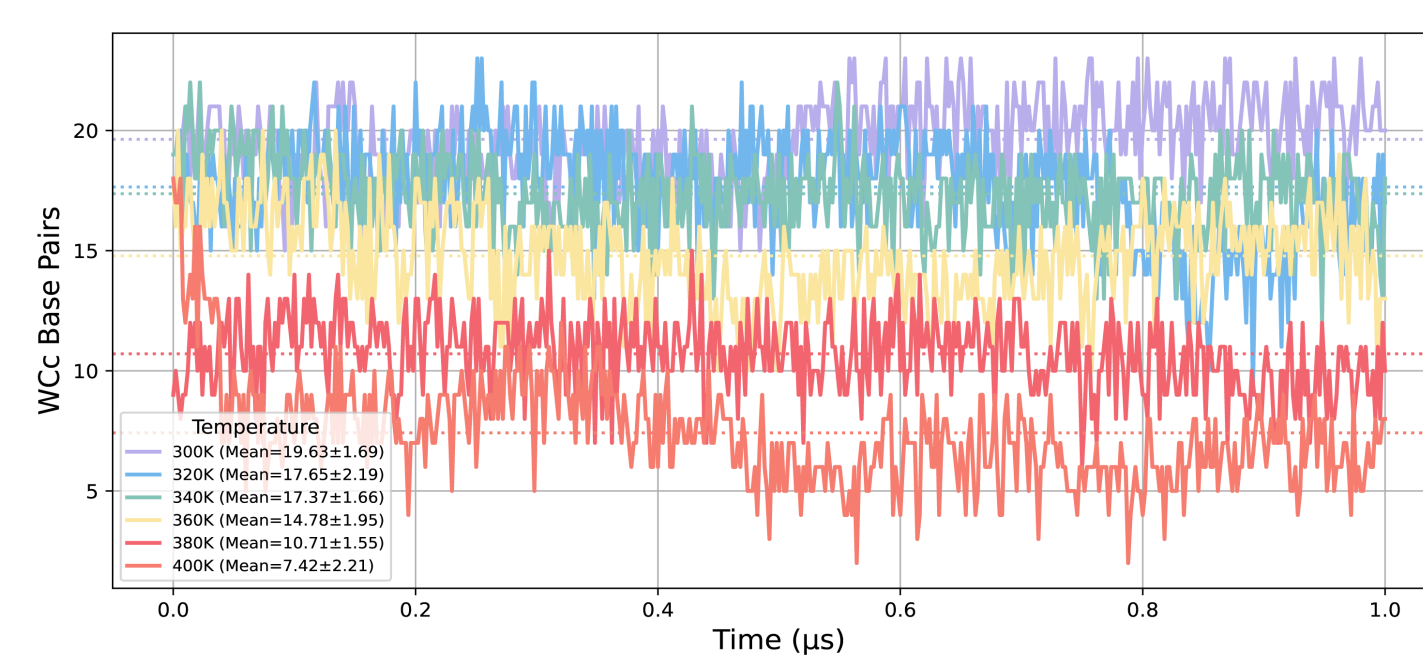


Fig 4.3. Temporal evolution of WcC base-pair counts for 1Y26 at 300–400 K over 1 μ s. Dashed lines indicate the mean count for each temperature.

- At **300K**, most sites **maintain** WcC pairing (Leontis–Westhof nomenclature, canonical cis Watson–Crick pairs: A–U, G–C), whereas at **360K** shows **frequent transitions** to WWc (Leontis–Westhof nomenclature, cis base pairs formed through the Watson–Crick edges; here shown as non-canonical WWc counts, i.e. excluding WcC) and sustained loss of pairing, indicating **partial melting**. The statistical comparison confirms **significantly fewer WcC base pairs** at 360K than at 300K, consistent with the **temperature-dependent decline** in total WcC counts over time across 300–400K, supporting **partial melting** of the base-pairing network at elevated temperatures.

FUTURE WORK

- In future studies, **Free Energy Perturbation (FEP)** methods will be employed to complement the current melting simulations. While melting analyses provide insights into structural stability and melting behavior at different temperatures, FEP allows for quantitative estimation of the **free energy differences** associated with **specific base modifications** or **sequence variants**. Integrating FEP into our workflow could help identify how chemical modifications influence RNA stability, thereby offering a more comprehensive understanding of the **thermodynamic contributions** underlying RNA folding and melting. This approach will provide predictive power for designing stable **mRNA sequences** and may **guide vaccine optimization**.

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